Year 7
Mathematics

Unit 3: Equations, Functions and Formulae

3.5 Brackets and Formulae

Brackets and Formulae

STARTER · 5 MIN

Date: _____ **Title: Brackets and Formulae**

LEARNING OBJECTIVES

- To be able to expand brackets

STARTER *Answer in your book. No calculators.*

1. Work out 3×14

2. Work out $3 \times 10 + 3 \times 4$

3. Work out $5 \times (8 - 2)$

4. Work out $5 \times 8 - 5 \times 2$

Starter — Answers

STARTER

Tick or correct in green pen

1. **42**

2. **42**

3. **30**

4. **30**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can expand a bracket with a positive number outside	☐
SILVER	I can expand a bracket with a negative number outside	☐
GOLD	I can expand brackets with a letter outside and use them in formulae	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Brackets and Formulae

INTRODUCTION · 3 MIN

Expand: multiply every term inside the bracket by the term outside.

Expand

Remove the bracket by multiplying.

$3(x + 4) = 3x + 12$

Every term

The outside term multiplies each one.

$5(y - 2) = 5y - 10$

Negative outside

Signs: $- \times + = -$, $- \times - = +$

$-2(x + 3) = -2x - 6$

Letter outside

$x \times x = x^2$

$x(x - 5) = x^2 - 5x$

Examples 1 and 2 — Expanding a Bracket

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Multiply every term $3(x + 4) = 3x + 12$

Example 1

Expand

$3(x + 4)$

Example 2

Expand

$5(y - 2)$

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

Multiply both terms by 3

$3(x + 4) = 3 \times x + 3 \times 4$

$= \mathbf{3x + 12}$

Tip: not $3x + 4$: multiply the 4 too.

Example 2

Multiply both terms by 5

$5(y - 2) = 5 \times y - 5 \times 2$

$= \mathbf{5y - 10}$

Tip: keep the − sign.

Task 1

PRACTICE · 9 MIN

BRONZE *Expand*

Hint Multiply every term $3(x + 4) = 3x + 12$

1. $2(x + 5)$
2. $4(a + 3)$
3. $3(y - 1)$
4. $6(m - 4)$
5. $2(3x + 1)$

★ $5(2p - 3)$

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. $2x + 10$
2. $4a + 12$
3. $3y - 3$
4. $6m - 24$
5. $6x + 2$

★ $10p - 15$

Examples 3 and 4 — Negative Numbers Outside

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint $- \times + = -$ $- \times - = +$

Example 3

Expand

$-2(x + 3)$

Example 4

Expand

$-3(y - 4)$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Multiply both terms by −2

$-2 \times x = -2x$

$-2 \times 3 = -6$

$\mathbf{-2(x + 3) = -2x - 6}$

Tip: negative × positive = negative.

Example 4

Multiply both terms by −3

$-3 \times y = -3y$

$-3 \times -4 = +12$

$\mathbf{-3(y - 4) = -3y + 12}$

Tip: negative × negative = positive.

Task 2

PRACTICE · 9 MIN

SILVER *Expand*

Hint − × + = − − × − = +

1. $-2(a + 4)$

2. $-5(x + 1)$

3. $-3(m - 2)$

4. $-4(y - 5)$

5. $-(x + 7)$

★ $-2(3b - 1)$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1 $-2a - 8$

2 $-5x - 5$

3 $-3m + 6$

4 $-4y + 20$

5 $-x - 7$

★ $-6b + 2$

Examples 5 and 6 — Brackets in Formulae

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

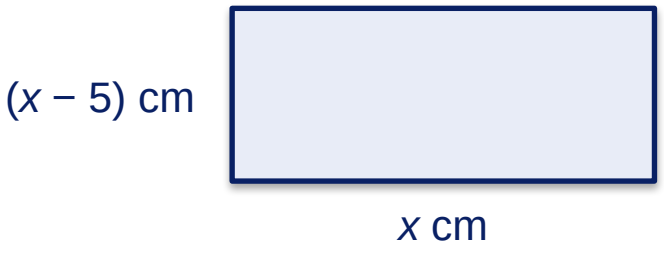
Hint $x \times x = x^2$ Expand, then collect

Example 5

(2 marks)

The rectangle has length x cm and width $(x - 5)$ cm.

Write and expand a formula for its area A.



Example 6

(2 marks)

Sam earns $\$x$ an hour, plus \$3 for each hour on a Sunday.

Write and expand a formula for his pay P for 2 Sunday hours.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Area = length × width

$A = x(x - 5)$

$= x^2 - 5x$

Tip: $x \times x = x^2$.

Example 6

2 hours at $(x + 3)$ dollars

$P = 2(x + 3)$

$= 2x + 6$

Tip: the bracket keeps $x + 3$ together.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: expand and simplify

Hint $x \times x = x^2$ Expand, then collect

1. $x(x + 2)$

2. $a(a - 7)$

3. $3(x + 2) + 4$

4. $2(y + 1) + 3(y + 2)$

5. Ella says $3(x + 4) = 3x + 4$.
   Explain her mistake.

★ Rectangle x by $(x + 3)$.
Expand its area.

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1 $x^2 + 2x$

2 $a^2 - 7a$

3 $3x + 10$

4 $5y + 8$

5 She did not multiply the 4 by 3: $3x + 12$.

★ $x^2 + 3x$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Expand $4(x + 3)$

SILVER Expand $-2(y - 5)$

GOLD Expand $x(x - 5)$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $4x + 12$

SILVER $-2y + 10$

GOLD $x^2 - 5x$

Exam Practice — Expanding Brackets

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Multiply every term $3(x + 4) = 3x + 12$

Question 1

(2 marks)

Expand

(a) $4(x + 2)$ (1)

.....

(b) $3(y - 5)$ (1)

.....

Question 2

(2 marks)

Expand

(a) $2(3a + 4)$ (1)

.....

(b) $5(2b - 1)$ (1)

.....

Exam Practice — Negatives and Mistakes

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint − × + = − − × − = +

Question 3

(2 marks)

Expand

(a) −2(*x* + 6) (1)

.....

(b) −4(*y* − 3) (1)

.....

Question 4

(2 marks)

Ella expands 3(*x* + 4) and gets 3*x* + 4.

Explain her mistake and give the correct answer.

.....

Exam Practice — Brackets in Formulae

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint $x \times x = x^2$ Expand, then collect

Question 5

(2 marks)

A rectangle is x cm long and $(x - 3)$ cm wide.

Write an expanded expression for its area.



.....

Question 6

(3 marks)

Expand and simplify

(a) $2(x + 4) + 3(x - 1)$ (2)

.....

(b) $a(a + 6)$ (1)

.....

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

(a) $4x + 8$ — B1

(b) $3y - 15$ — B1

Tip: multiply every term.

Question 2

(a) $6a + 8$ — B1

(b) $10b - 5$ — B1

Tip: $2 \times 3a = 6a$.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

(a) $-2x - 12$ — B1

(b) $-4y + 12$ — B1

Tip: − × − = +.

Question 4

She did not multiply 4 by 3 — B1

$3x + 12$ — B1

Tip: the 3 multiplies both terms.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

Area $= x(x - 3)$ — M1

$= x^2 - 3x$ — A1

Tip: $x \times x = x^2$.

Question 6

(a) $2x + 8 + 3x - 3 = 5x + 5$ — M1 A1

(b) $a^2 + 6a$ — B1

Tip: expand both, then collect.